

CS 471: Final Project Grading Rubric

Total: 250 Points — U.S. Air Force Academy

Names: _____

Section: _____

Milestone 1: Project Design Worksheet (25 pts)

Lesson 32 Focus: Defining a mathematically feasible, well-scoped Minimum Viable Product (MVP) using one core algorithm.

Criteria	Exemplary Target (100%)	Deficient / Penalty Trigger	Pts
Problem & Environment	Clear tactical challenge. States, actions, observations, and rewards are mathematically defined.	Vague problem, lacks military context, or missing state/reward definitions.	10
Algorithmic Scope	Selects exactly one core algorithm. Scope is realistic and isolated for a 4-lesson build.	Multi-algorithm mashup, unfeasible scope, or wrong algorithm for domain.	10
Evaluation Metric	Defines precise mathematical metrics (e.g., loss convergence, PR curves) to prove learning.	Metrics are purely qualitative, subjective, or missing entirely.	5

Milestone 2: Functional Checkpoint (50 pts)

Lesson 36 Focus: Demonstrating execution of core mathematical logic without crashing (terminal-only output is acceptable).

Criteria	Exemplary Target (100%)	Deficient / Penalty Trigger	Pts
Execution & Stability	Core logic executes end-to-end without runtime errors (e.g., search completes, Q-table updates).	Code crashes, enters infinite loops, or fails to run during evaluation.	30
Logic Verification	Terminal/log output mathematically proves algorithmic state transitions and inferences are active.	Logic is hollow, hardcoded, or acts as a random baseline without updates.	20

Milestone 3: Final Technical Deliverable (150 pts)

Lesson 41 Focus: Complete agent implementation, quantitative validation, operational impact analysis, and code integrity.

Criteria	Exemplary Target (100%)	Deficient / Penalty Trigger	Pts
Algorithm & Math Integrity	Algorithm implemented with mathematical rigor. Efficiently handles edge cases without black-box shortcuts.	Major mathematical flaws, inadmissible heuristics, or improper logic updates.	45
Proof of Learning	Rigorously proves learning/optimization via empirical data, visualizations, and statistical context vs a baseline.	Fails to prove learning; relies on anecdotal runs, no graphs, or no improvement.	40
Tactical Analysis	Insightful analysis connecting behavior to tactical outcomes. Evaluates failure modes, tradeoffs, and risks.	Superficial operational analysis; treated purely as an abstract coding exercise.	35
Code & Docs	Modular, clean, completely reproducible. Any AI-assisted code is verified and defensible.	Disorganized, undocumented codebase, or contains unverified AI generation.	30

Milestone 4: Presentation & Oral Defense (25 pts)

Lesson 41 Focus: Communicating system design, defending engineering decisions, and verifying codebase mastery.

Criteria	Exemplary Target (100%)	Deficient / Penalty Trigger	Pts
Operational Brief	Professional demonstration. Clearly articulates architecture, mathematical metrics, and tactical impact.	Disorganized, runs over time, or fails to show working agent and data.	15
Logic Defense	Every member shows deep, granular understanding of the codebase and can independently explain any tradeoff.	Cannot explain submitted code; reliance on peer carry or unverified AI logic.	10

Individual Accountability & Contribution Policy

- **Team Structure:** Teams consist of 2–3 students.
- **Peer & Instructor Evaluation:** Final grades are subject to an **Individual Multiplier (0.0 to 1.0)** based on instructor observation and mandatory end-of-project peer contribution reviews.
- **Non-Contribution Penalty:** Failure to meaningfully contribute to the codebase, design milestones, or defense will result in an individual point reduction, up to and including a zero on the final deliverable.